Saheed Faremi

PhD researcher, University College Cork · Data scientist
saheedfaremi@gmail.com · Dublin, Ireland
GitHub · LinkedIn · Google Scholar · ORCID

PhD researcher and data scientist with five years of experience building production software. I study what brief brain states represent and how they change over time. In industry, I apply machine learning to healthcare, finance and compliance.
Last updated November 2026

RESEARCH AREAS

Brain and behaviour: EEG microstates and brain-signal analysis; neuroscientific knowledge discovery; neuroimaging; nonlinear time-series analysis

Explainable and reproducible AI: Explainable Artificial Intelligence; mechanistic interpretability; reproducible machine learning

Applications: Deep learning for health and behaviour; production AI and data systems

TECHNICAL SKILLS

Brain-signal analysis: PyTorch; scikit-learn; MNE; large-scale model evaluation

Applied AI and document automation: Azure OpenAI; document question answering; fraud and anomaly detection; model fine-tuning; document generation and automation

Software & cloud engineering: Python; Go; TypeScript; SQL; FastAPI; Azure; CI/CD; SLURM; automated testing

Research evaluation: Testing across participants; statistical significance tests; correction for multiple comparisons; time-series and duration analysis

EDUCATION

PhD, University College Cork In progress
Neural engineering and machine learning. Supervisor: Luca Longo.

MSc Computer Science, Technological University Dublin Completed 2025
Machine learning, deep learning, data mining, visualisation and statistics. 4.0/4.0 GPA (80/100).

BSc Information Technology, University of Eswatini Completed
Second Class Honours, Upper Division (2.1).

EXPERIENCE

PhD researcher In progress
Artificial Intelligence and Cognitive Load Research Lab (AICL), University College Cork

- Developed and compared image-based scalp maps with maps built from connections between recording electrodes. When both used the same processing step on 203 resting-state recordings, the electrode-based approach accounted for 61.8% of the signal versus 54.2% for the image-based approach. The image-based approach produced tighter groups of activity.
- Tested whether including the duration of a brain state improves the model's match to the recording. With nine states, including duration improved the match from 55.7% to 63.7%, compared with using only the current state. A separate comparison found that its state-switching patterns differed more from chance than those from standard clustering (35.6% versus 23.6%).
- Showed that processing choices can change the reported duration of brain states. The gap between the two models narrowed from 70.2 versus 143.2 milliseconds to 70.2 versus 71.0 milliseconds when both used the same processing step. In a separate test, changing a duration setting moved the average from 402 to 24 milliseconds.

Data scientist to Founding engineer May 2021 to present
Etihuku (Curnance)

- Delivered seven production services for payments, account security, identity checks and background processing, plus a 120-screen mobile app, an admin dashboard and a public website.
- Fixed a live withdrawal-fraud issue by debiting the customer's account before making an external payout and rejecting duplicate transfer requests.
- Built a fraud and anti-money-laundering system with 38 rule-based checks, AI-drafted suspicious-activity reports, and privacy controls that hide personal details.
- Built an Azure AI document service that cut manual document work by 85% across three compliance regions.

PUBLICATIONS

Peer-reviewed

1. Temporal Dynamics and Latent-Space Interpretability of Generative EEG Microstate Segmentation.
   Saheed Faremi, Andrea Visentin, Luca Longo.
   AICS 2026, University College Dublin (accepted; proceedings to appear), 2026.
2. Integrating Convolutional Variational Autoencoders and the Gaussian Mixture Model for efficient manifold learning and clustering of spatially preserved EEG topographic maps.
   Saheed Faremi, Luca Longo.
   Brain Informatics, 2026. DOI: 10.1186/s40708-026-00327-9.
3. Machine Learning Models for Identifying Factors Influencing and Predicting Malaria Among Children Under Five Years in Nigeria.
   Akinpelumi Saheed Faremi, Boluwaji Akinnuwesi, Elliot Mbunge, Petros M. Mashwama, Stephen Fashoto, Polite Zenzo Ncube, John Batani, Shamsudeen Ademola Sanni, Yinusa A. Faremi, Andile Metfula.
   IEEE ICTAS 2024, 2024. DOI: 10.1109/ICTAS59620.2024.10507142.

Preprints, registered protocols & doctoral work

4. Interpretable EEG Microstate Discovery via Variational Deep Embedding: A Systematic Architecture Search with Multi-Quadrant Evaluation.
   Saheed Faremi, Andrea Visentin, Luca Longo.
   XAI 2026, Late-breaking work and Doctoral Consortium track, Fortaleza, Brazil (arXiv preprint), 2026. DOI: 10.48550/arXiv.2605.10947.
5. Autoencoder-Based Models for Scalp EEG: A Systematic Review of Architectures, Applications, Latent Representations, Interpretability, and Validation.
   Morteza Akbari, **Saheed Akinpelumi Faremi**, Luca Longo.
   OSF Registries (registered systematic-review protocol), 2026. DOI: 10.17605/osf.io/v2w3z.
6. Explainable Disentangled Representation Learning of Recurring Brain Activation Patterns via Variational Autoencoders.
   Saheed Faremi.
   XAI World Conference 2025, Doctoral Proposals track, 2025.

Under review and manuscripts

7. Convolutional Variational Deep Embedding Produces Canonical EEG Microstate Maps and Richer Transition Dynamics than Modified K-Means on Resting EEG.
   Saheed Faremi, Andrea Visentin, Luca Longo.
   Cognitive Computation (manuscript in preparation), 2026.
8. Spatial Encoding Geometry Determines the Maps and Transitions Recovered by Deep EEG Microstate Models.
   Saheed Faremi et al.
   ICLR 2027 (under review; OpenReview #50003), 2026.
9. Prediction of Early Detection of Autism Spectrum Disorder in Toddlers Using Explainable Artificial Intelligence.
   Stephen Fashoto et al.
   Under review, 2026.

TALKS

- Temporal Dynamics and Latent-Space Interpretability of Generative EEG Microstate Segmentation.
  34th Irish Conference on Artificial Intelligence and Cognitive Science (AICS 2026), University College Dublin, 2026.
- Interpretable EEG Microstate Discovery via Variational Deep Embedding.
  XAI 2026, Late-breaking work and Doctoral Consortium track, Fortaleza, Brazil, 2026.
- Machine Learning Models for Predicting Malaria in Nigerian Children Under Five.
  IEEE ICTAS 2024, 2024.